RoamScope | Teammate Test Guide

Test package: RoamScope_Teammate_Test_v4.zip | macOS pilot | Updated 19 October 2026

Purpose. Verify that the macOS helper exposes SSID/BSSID to RoamScope and captures display RSSI, channel, band and primary-channel center frequency. Validate real AP-to-AP roaming separately.

1. Prerequisites

Use a Mac with Wi-Fi enabled. Python 3 and Apple Command Line Tools / Swift are needed. Homebrew is **not** included with macOS by default. Locally built apps and Location Services must be permitted by your organization; do not bypass corporate policy.

2. Install Python 3 using Homebrew (if needed)

Check whether Homebrew and Python 3 are already available:

```
brew --version
python3 --version
```

If **brew** works but **python3** is missing, or the team wants a Homebrew-managed Python, run:

```
brew install python
python3 --version
```

If Homebrew is missing, use your organization-approved software channel to obtain it, or ask IT. Do not install Homebrew on a managed Mac without approval. Homebrew reference: https://brew.sh/

Check for Apple Command Line Tools (needed to build the Swift helper):

```
xcode-select -p
xcrun --find swiftc
```

If unavailable, contact IT for an approved installation of Command Line Tools.

3. Extract and prepare RoamScope

Extract the ZIP so that roamscope.py is at ~/Downloads/roamscope/roamscope.py, then run:

```
cd ~/Downloads/roamscope
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

If shell activation is unreliable, use `./.venv/bin/python` explicitly. An old (.venv) prompt alone does not prove that the environment works.

4. Build and open the macOS Wi-Fi helper

```
cd ~/Downloads/roamscope/RoamScopeHelper
bash ./build.sh
open ./RoamScopeWiFiHelper.app
```

In the helper, select **Request Location Access** if needed, then **Read Wi-Fi**. Confirm SSID and BSSID are populated; keep the helper open throughout the capture. If blocked by enterprise policy, contact IT.

5. Check the bridge and doctor

```
curl --noproxy '*' --max-time 5 http://127.0.0.1:18763/
cd ~/Downloads/roamscope
source .venv/bin/activate
python roamscope.py --doctor
```

Expected: JSON and doctor output with a valid BSSID. The CoreLocation authorization fields may differ from BSSID availability. If missing, check the helper and listening process:

```
lsof -nP -iTCP:18763 -sTCP:LISTEN
```

6. Capture and inspect results

```
python roamscope.py --duration 300 --interval 2 --csv
```

Collect for five minutes, moving between known AP coverage areas where permitted. Then open `roamscope-out/report.html`. Review Access points seen, band/frequency and Roam events. A stable BSSID with zero roams does not validate roaming detection.

7. Send results back

Send the roamscope-out folder (HTML report, SQLite database and CSV exports), a helper screenshot, doctor output, macOS version and any build errors. Review SSIDs/BSSIDs for sensitivity before sharing externally.

Validation status

Python compilation and simulated capture were checked for v4. Native Swift compilation, permissions and real AP-to-AP roaming require testing on the teammate Mac. This is a pilot build, not a production-certified release.